Short Biography

Moez A. AbdelGawad is an Assistant Professor of Quantum Informatics and Programming Languages at the Informatics Research Institute (IRI), SRTA-City. Since 2020, he has continuously also been a remote Visiting Scientist at the Computer Science Department, Rice University, and an Adjunct Assistant Professor at the Computer & Systems Engineering Department, Faculty of Engineering, Alexandria University.

The current research of Dr. AbdelGawad focuses on identifying efficient quantum algorithms for classically intractable problems, drawing on quantum informatics and tools from spectral and algebraic graph theory. To support researchers and students in the field, he recently introduced QAL (Quantum Algorithms Lab), an interactive web application for researching and teaching quantum algorithms, and has proposed establishing a Quantum Informatics Lab (QI Lab) at IRI, SRTA-City.

Dr. AbdelGawad earned his Ph.D. in Computer Science from Rice University in 2011. His dissertation introduced NOOP, a precise mathematical model of nominally-typed object-oriented programming, establishing that inheritance and subtyping are effectively identified in mainstream OOP languages — work he later adapted into a book (2013). He has since published on the semantics of generic, nominally-typed OOP languages such as Java and C#, including work presented at the Applied Category Theory Conference at Oxford University in 2019.

Dr. AbdelGawad was a Visiting Research Scholar at the Category Theory Group, Hunan University, China (2015–2016), and earlier interned with the Programming Languages Group at EPFL, Switzerland, working on the Scala compiler (2005). He holds an M.Sc. and B.Sc. in Computer Science from Alexandria University, with theses on a novel encryption mode (2DEM) and automated human face recognition, respectively.

Dr. AbdelGawad has a long-standing interest in STEM education and public outreach — including authoring the first Arabic-language instructional videos on solving Rubik's Cube. Further details on his research, teaching, and publications are available online in his CV and his extended biography.